Project Title: Adaptive Traffic Signal Control Using RL

Group name: Rush Hour
Group 32
Team members:

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Problem statement:
Traffic congestion increases travel time, fuel consumption, and fuel emission. Traditional traffic lights fail to adapt dynamically to real time conditions. Therefore, there is a need for an intelligent traffic signal control system that optimizes signal timings in real time to minimize congestion and improve overall traffic efficiency. In addition, how can a model be trained using historical traffic patterns corresponding to specific time period (like evening rush hour peak or special events).

Related work:

- Wei, Hua, Guanjie Zheng, Huaxiu Yao, and Zhenhui Li. 2019. *IntelliLight: A Reinforcement Learning Approach for Intelligent Traffic Light Control.* In **Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '19)**. Association for Computing Machinery, New York, NY, USA, 2496–2505. https://dl.acm.org/doi/10.1145/3219819.3220096
- Van der Pol, Elise, and Frans A. Oliehoek. 2016. Coordinated Deep Reinforcement Learners for Traffic Light Control. In *Proceedings of the 30th International Conference on Neural Information Processing Systems (NeurIPS '16)*. Curran Associates Inc., Red Hook, NY, USA, 3474–3482. https://www.elisevanderpol.nl/papers/vanderpolNIPSMALIC2016.pdf

Initial hypothesis:
We aim to investigate how different RL models perform in adaptive traffic signal control under complex scenarios. We believe advance policy-based methods will demonstrate greater

robustness and scalability compared to value-based methods when applied to more complex scenarios.

Dataset(s):

We are planning to use the Traffic Signal Control Benchmark Hangzhou 1x1 traffic scenario dataset. It contains roadnet.json (road network structure) & flow.json (vehicle flow definition).

- Training Set: hangzhou_1x1_bc-tyc_18041607_1h (April 16, 2018, 7:00 AM – 8:00 AM)
- Validation Set: hangzhou_1x1_bc-tyc_18041608_1h (April 16, 2018, 8:00 AM – 9:00 AM) & hangzhou_1x1_bc-tyc_18041610_1h (April 16, 2018, 9:00 AM – 10:00 AM)
- Test Set: hangzhou_1x1_kn-hz_* , hangzhou_1x1_qc-yn_* & hangzhou_1x1_sb-sx_*

We will check generalization across time (same intersection, different time) and generalization across location (unseen intersection).

This dataset was in CityFlow format, but we will be using SUMO due to better visualization and more industry relevance, so we will have to convert each intersection to SUMO format which is .net.xml (Network file), .rou.xml (Route file) & .sumocfg (SUMO config file). We will also have to verify traffic signal phases and vehicle flow and fix random seeds for reproducibility.

For each intersection, roadnet.json is around 33kb and flow.json is approx. 700kb.

Dataset source (link and reference)	Link Zhang et al., *CityFlow: A Multi-Agent Reinforcement Learning Environment for Large Scale City Traffic Scenario*, KDD 2019.
Number of instances	Approx. 720 decision steps per episode (1 hr of traffic data)
Number of features	15-20 depending on number of lanes and traffic signal phases
Class distribution (# instances in each class, if applicable)	Not applicable
Dataset splits	• Training Set: hangzhou_1x1_bc-tyc_18041607_1h • Validation Set: hangzhou_1x1_bc-tyc_18041608_1h & hangzhou_1x1_bc-tyc_18041610_1h • Test Set: hangzhou_1x1_kn-hz_* , hangzhou_1x1_qc-yn_* & hangzhou_1x1_sb-sx_*

Preprocessing steps	• Converting dataset format from CityFlow to SUMO • Verify Traffic Signal phases and vehicle flow. • Fix random seeds

Method(s):
We will use multiple RL algos including PPO, DQN and A2C. These algorithms represent both value-based and policy-based approach allowing a comparison between different learning methods. The novelty lies in first benchmarking different learning methods on single real world intersection before scaling the best performing model to more complex network. Additionally, the study leverages the study of historical traffic patterns corresponding to specific time periods (eg. morning peak or evening rush hour) enabling the development of control policies based on anticipated traffic conditions. While other work such as IntelliLight focusses on optimizing a single algorithm for a fixed intersection and multi agent approach on large scale networks, our study introduces a structured progression from controlled single intersection to scalability analysis. We will use SUMO for simulation and Stable Baselines3 for RL. Compared to the state of the art multi agent method used in CityFlow benchmark, this work focusses on controlled single agent benchmarking and then generalizing across time and intersection. Additionally, this work will include time variant traffic control by training and evaluating models across different time periods and cross intersection generalization.

Evaluation:
We will quantitatively measure the performance using traffic efficiency metrics using the simulator like average waiting time, average queue length, average travel time, and throughput. We will do a qualitative assessment by visually inspecting the simulator to observe whether the traffic flow appears to be smoother. To compare our method to existing method, we will benchmark PPO/DNQ/A2C against baseline controllers like fixed time signal and a defined policy and report mean and variance of different metrics to show stability and robustness. We will also evaluate generalization by training on one time frame and testing on different time frame and unseen intersections measuring how well each method maintains performance under changing traffic patterns.

Management plan:
We plan to begin with one week of data preprocessing (converting data to be suitable for our simulator) followed by one week to define all the control policies and reward functions for our models (DNQ, PPO, A2C) and then next two weeks for training and testing different models on a single intersection. And Then Two weeks to train and test our best model for more complex traffic intersections. We will keep 1 week for any unseen problem and the remaining time for creating report and presentation which matches the course timeline. Each group member will take responsibility for implementing and evaluating one model to ensure parallel progress and balance workload. Then each group member will refine policies and test on more complex intersection to find the best policies for more complex traffic control. Progress will be monitored through weekly meetings and shared documents with code review and result comparison to maintain accountability and collaboration.